Self-Powered Weather Monitoring System

A project report submitted on Design Thinking and Innovation in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (Internet of Things)

Submitted by

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under the esteemed guidance of

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Professor

[Program: Computer Science and Engineering (Internet of Things) – CSO]

VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

(Autonomous)

Approved by AICTE, Permanently Affiliated to JNTUK, NAAC Accredited with 'A' Grade, ISO 9001:2015 Certified

Nambur (V), Pedakakani (M), Guntur (Dt.), Andhra Pradesh – 522 508

2025

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CERTIFICATE

This is to certify that the project report entitled "**Self-Powered weather monitoring System"** is being submitted by **M. Bhargavi (**Regd. No**:23BQ1A4929**), **N. Vennela (**Regd. No**:23BQ1A4934**), **P. V. R. Lakshmi (**Regd. No**: 23BQ1A4939**) and **T. Pallavi (**Regd. No**:23BQ1A4948**) on Design Thinking and Innovation in partial fulfillment of the requirement for the award of the degree of the **Bachelor of Technology** in **Computer Science and Engineering (Internet of Things)** to the Vasireddy Venkatadri Institute of Technology is a record of bonafide work carried out by them under my guidance and supervision.

The results embodied in this project have not been submitted to any other university or institute for the award of any degree or diploma.

Signature of the Supervisor	**Head of the Department**
Dr. S. Krishna Prasad	Dr. Chintalapudi V Suresh
Professor,	Professor & HoD,
Department of CSO, VVIT.	Department of CSO, VVIT.

DECLARATION

We hereby declare that the work embodied in this research project entitled "**Self-Powered Weather Monitoring System**", which is being submitted by us in requirement for the B. Tech Degree in **Computer Science and Engineering (Internet of Things)** from Vasireddy Venkatadri Institute of Technology, is the result of investigations carried out by us under the supervision of Dr. S. Krishna Prasad, Professor.

The work is original and the results in this project have not been submitted elsewhere for the award of any degree or diploma.

Signature of the Candidates

M. Bhargavi (Regd. No**: 23BQ1A4929**)

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ABSTRACT

Human beings, infrastructure, and the environment are critically endangered by severe weather patterns. This project aims to develop an IoT-based weather monitoring system that extreme weather events in real time. The system includes a variety of environmental sensors, including temperature, humidity, barometric pressure, rainfall, among others that are interfaced to a low-power microcontroller with a wireless link.

System powered by renewable energy sources, primarily solar energy, with efficient power management circuitry for uninterrupted operation enables remote or off-grid location. The data collected is sent to cloud platform for remote monitoring, analysis. This project focuses on maintenance free, scalable, resilient solution unsuitable for deployment in vulnerable and isolated regions. This selfpowered IoT system is capable of providing climate research, smart environmental management capabilities, and thus, supporting initiatives.

Department Vision

To accomplish the aspirations of emerging engineers to attain global intelligence by obtaining computing and design abilities through communication that elevate them to meet the needs of industry, economy, society, environmental and global.

Department Mission

- To mould the fresh minds into highly competent IoT application developers by enhancing their knowledge and skills in diverse hardware and software design aspects for covering technologies and multi-disciplinary engineering practices.
- To provide the sate- of- the art facilities to forge the students in industry-ready in IoT system development.
- To nurture the sense of creativity and innovation to adopt the socio-economic related activities.
- To promote collaboration with the institutes of national and international repute with a view to have best careers.
- To enable graduates to emerge as independent entrepreneurs and future leaders.

Program Educational Objectives (PEOs)

PEO-1: To formulate the engineering practitioners to solve industry's technological problems

PEO-2: To engage the engineering professionals in technology development, deployment and engineering system implementation

PEO-3: To instil professional ethics, values, social awareness and responsibility to emerging technology leaders

PEO-4: To facilitate interaction between students and peers in other disciplines of industry and society that contribute to the economic growth**.**

PEO-5: To provide the technocrats the amicable environment for the successful pursuing of engineering and management.

PEO-6: To create right path to pursue their careers in teaching, research and innovation.

Program Outcomes (POs)

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO-1: Proficient and innovative with a strong cognizance in the arenas of sensors, IoT, data science, controllers and signal processing through the application of acquired knowledge and skills.

PSO-2: Apply cutting-edge techniques and tools of sensing and computation to solve multi-disciplinary challenges in industry and society.

PSO-3: Exhibit independent and collaborative research with strategic planning while demonstrating professional and ethical responsibilities of the engineering profession.

Project Outcomes

Students who complete a minor project will:

PW-01. Use the design principles and develop concept for the project

PW-02. Estimate the time frame and cost for the project execution and completion.

PW-03. Analyze the project progress with remedial measures individual in a team.

PW-04. Examine the environmental impact of the project.

PW-05. Demonstrate the project functionality along with report and presentation.

PW-06. Apply the Engineering knowledge in design and economically manufacturing of components to support the society need.

PW-07. Assess health, safety and legal relevant to professional engineering practices.

PW-08. Comply the environmental needs and sustainable development.

PW-09. Justify ethical principles in engineering practices.

PW-010. Perform multi-disciplinary task as an individual and / or team member to manage the project/task.

PW-011. Comprehend the Engineering activities with effective presentation and report.

PW-012. Interpret the findings with appropriate technological / research citation.

MAPPING OF PROJECT OUTCOMES TO POs

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PW-01	3	2	2	2								
PW-02	3	2	2								3	
PW-03	3	3		2	3					3		
PW-04	3					3	3	3				3
PW-05	3	2									3	
PW-06	3	2	2	2	3							
PW-07						3						
PW-08							3					
PW-09								3				
PW-10									3		3	
PW-11										3		
PW-12												3
PW-PO	**3**	**2**	**2**	**2**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**

MAPPING OF PROJECT OUTCOMES TO PSOs

	PSO1	PSO2	PSO3
PW-01	2	2	2
PW-02			2
PW-03			
PW-04	3	3	3
PW-05			
PW-06	2	2	2
PW-07	2	2	2
PW-08	1	1	1
PW-09	2	2	2

PW-10	2	2	2
PW-11	2	2	2
PW-12	1	1	1
PW-PSO	**2**	**2**	**2**

Note: Strong – 3, Moderate – 2, Low – 1

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CHAPTER 1

INTRODUCTION

INTRODUCTION: -

A weather station is a device that can monitor various parameters like humidity, temperature, pressure, rainfall, etc... of the surrounding environment." Solar-Powered Wi-Fi Weather Station Using NodeMCU and ThingSpeak " offering a sustainable and efficient method for monitoring atmospheric conditions while leveraging cloud-based data management. As the demand for accurate and real-time weather data continues to grow, the integration of renewable energy sources with Internet of Things (IoT) technology has paved the way for innovative solutions.

"THE SOLAR POWERED WI-FI WEATHER STATION PROJECT" not only used in an educational tool for understanding IoT and renewable energy but also contributes to the growing field of smart agriculture, environmental monitoring, and climate research. By building this project, developers can gain experience and learning in electronics, programming, and data analysis.

The ESP8266 Wi-Fi module serves as the foundation for the NodeMCU, which enables smooth internet access and transmits the gathered data to the ThingSpeak platform.The weather station runs independently, which lessens dependency on conventional energy sources and encourages sustainability. The Solar Powered Wi-Fi Weather Station using NodeMCU and ThingSpeak is not just a technological, It is a useful instrument that explains sustainable development and smart agriculture.

Solar weather monitoring systems have a wide range of applications. In the management of renewable energy, solar weather monitoring systems ensure maximum solar panel efficiency and energy generation. In water management and agriculture, weather monitoring systems helps farmers and water managers in decision-making on crop management, irrigation, and water availability. Solar weather monitoring systems also furnish vital data for disaster recovery and response

Table:1 Components required

S. No	Name of the component	Specifications	Number of Units required
1	ESP8266	NodeMCU	1
2	DHT11 Sensor	Temperature and Humidity sensor	1
3.	BMP280 Sensor	Pressure sensor	1
4	Rain sensor	Rain detector	1
5	Li-ion charger	TP4056	1
6	Li-ion Battery	3.7V 18650	1
7	Step-Up Booster	DC-DC 0.9V-5V	1
8	Solar Panel	6V, 100Ma	1
9	Jumper Wires	Connecting the components	required

1.2 OVERVIEW:-

A self-powered weather monitoring system collects data and environmental conditions like temperature, humidity, and pressure using sensors, often powered by solar panels and batteries. The Solar Powered Wi-Fi Weather Station is an innovative project designed to monitor and report environmental conditions in real-time, utilizing the NodeMCU microcontroller and the ThingSpeak IoT platform.

Key Components: -

- NodeMCU ESP8266: The main microcontroller that connects to Wi-Fi and processes data.
- DHT22 Sensor: Measures temperature and humidity with high accuracy.
- BMP180 Sensor: Measures atmospheric pressure and altitude.
- Rain Sensor: Detects rainfall to provide additional weather data.
- Solar Panel: Provides renewable energy to power the system.
- Battery: A 1000mAh Li-Ion battery stores energy for continuous operation.

Functionality: -

- Data Collection: The weather station collects data from various sensors at regular intervals.
- Data Transmission: The collected data is sent to ThingSpeak via Wi-Fi, allowing real-time monitoring.
- Deep Sleep Mode: The NodeMCU can enter a low-power state to conserve battery life, waking up periodically to take measurements.

Benefits:

- Renewable Energy Source: Lowers dependence on conventional energy sources and reduces damage to the environment.
- Enhanced Accuracy and Consistency: Delivers correct and consistent information to make improved decisions.
- Improved Public Safety: Accurate warnings and predictions save lives and property from dangerous weather.

1.3 LITERATURE

Solar weather monitoring systems have attracted a lot of attention in recent years based on the potential to deliver reliable and precise weather information while supporting sustainable use of energy. This literature review synthesizes existing knowledge about solar weather monitoring systems, their uses, and advantages.

The historical evolution of weather monitoring systems has progressed from the conventional manual observations and mechanical devices to more advanced electronic

sensors, wireless technology, and data analysis. The addition of solar power has made the systems more energy-efficient and sustainable.

Future opportunities for solar weather monitoring systems involve their integration with new technologies, such as artificial intelligence and the Internet of Things (IoT), that will potentially improve their functionality and areas of application. Their implementation will rise across different industries, such as renewable energy, agriculture, and emergency management.

1.4 PROBLEM IDENTIFICATION

The primary problem identification for a self-powered weather monitoring system is all about providing secure and sustainable power for operation, especially in distant areas, and ensuring data integrity and accuracy in harsh environments.

1. Power Supply and Sustainability:

Remote Location Challenges:

Remote locations usually do not have access to stable power grids, and it becomes challenging to provide a constant power supply to the weather monitoring system.

Solar Power Limitations:

Although solar power is a popular solution, its efficiency can be hampered by conditions such as cloudy weather, seasonal changes, and harsh weather conditions.

Energy Storage:

Effective energy storage systems (e.g., batteries) are essential in order to buffer power when there is low solar irradiance or the system is not in direct sunlight.

Maintenance and Replacement:

Routine maintenance of solar panels and battery systems, and timely replacement of hardware, is necessary for reliability over an extended time.

2. Data Accuracy and Integrity:

Sensor Quality and Calibration:

The reliability of weather information is based on the sensors' quality and calibration within the monitoring system.

Environmental Factors:

Severe weather conditions (e.g., extreme temperatures, wind, rain, dust) can impair sensor operation and produce false readings.

Data Transmission:

Secure and reliable data transmission to a central point is essential to provide timely and accurate weather data.

Data Security:

Wireless communication utilized by numerous systems can be subject to manipulation or interception and may need effective security controls.

CHAPTER 2

Project Description

2.1 PROJECT INTRODUCTION

This project aims to design and develop a solar-powered weather monitoring system that provides accurate and reliable weather data, promoting sustainable energy practices and enhancing our understanding of weather patterns. Weather monitoring systems are essential for various applications, including meteorology, agriculture, transportation, and emergency management. Traditional weather monitoring systems often rely on non-renewable energy sources, contributing to environmental concerns

Building a solar weather monitoring system involves several components and steps. Designing a solar weather monitoring system using BMP280, DHT11, and rain sensor with NodeMCU and ThingSpeak includes various components. The system includes a solar panel to supply power, BMP280 to sense atmospheric pressure and temperature, DHT11 to sense humidity and temperature, a rain sensor to sense precipitation, NodeMCU to process sensor data and send it to ThingSpeak, and ThingSpeak, a cloud-based data storage and analysis platform.

Then the components are selected, which include the solar panel, BMP280, DHT11, rain sensor, NodeMCU, and ThingSpeak. Then the components are assembled by interconnecting them. The NodeMCU needs to be coded to read data from sensors and send it to ThingSpeak. Last but not least, the system needs to be tested to validate that it is functioning properly.

Several considerations must be considered when constructing a solar weather monitoring system. Power management is essential to provide the system with a power source and ensure power management is efficient.The data transmission technique also needs to be selected, like Wi-Fi.A code example is provided to demonstrate how to process data from the sensors and transmit it to ThingSpeak. The code uses libraries from Adafruit to work with the BMP280 and DHT11 sensors. Resources are also provided, including NodeMCU, ThingSpeak, and Adafruit, to help with building the system.

Generally, designing a solar weather monitoring system with BMP280, DHT11, and rain sensor using NodeMCU and ThingSpeak is a good project for anyone interested in IoT and weather monitoring.

2.2 COMPONENTS USED

2.2.1 ESP8266

The NodeMCU (Node MicroController Unit) is an open-source hardware and software development platform based on a cheap System-on-a-Chip (SoC) by the name ESP8266. The ESP8266, which was designed and produced by Espressif Systems, has in it all the key components of a computer: CPU, RAM, networking (WiFi), and even a contemporary operating system and SDK. That makes it a great option for Internet of Things (IoT) projects of any sort. Still, as a chip, the ESP8266 is also difficult to access and make use of. You need to solder on wires, with the proper analog voltage, to its pins in order to accomplish simple tasks like turning it on or sending a keystroke to the "computer" on the chip. You need to also program it using low-level machine instructions that the chip hardware can understand. This integration on this level is not an issue with the use of the ESP8266 as an embedded controller chip in commercially mass-produced devices. It's a massive handicap for hobbyists, hackers, or students interested in trying to use it within their own IoT projects.

The NodeMCU ESP8266 is a popular, low-cost, open-source development board based on the ESP8266 Wi-Fi chip, commonly used for Internet of Things (IoT) applications, allowing easy programming with Arduino IDE and Lua. Control Pins are used to control the NodeMCU/ESP8266. These pins include Chip Enable pin (EN), Reset pin (RST) and WAKE pin. EN: The ESP8266 chip is enabled when EN pin is pulled HIGH. When pulled LOW the chip works at minimum power.

GPIO Pins NodeMCU/ESP8266 has 17 GPIO pins which can be assigned to functions such as I2C, I2S, UART, PWM, IR Remote Control, LED Light and Button programmatically. Each digital enabled GPIO can be configured to internal pull-up or pull-down, or set to high impedance. When configured as an input, it can also be set to edge-trigger or level-trigger to generate CPU interrupts.

FIG 2.1 ESP8266 Board

ADC Channel The NodeMCU is embedded with a 10-bit precision SAR ADC. The two functions can be implemented using ADC. Testing power supply voltage of VDD3P3 pin and testing input voltage of TOUT pin. However, they cannot be implemented at the same time.

UART Pins NodeMCU/ESP8266 has 2 UART interfaces (UART0 and UART1) which provide asynchronous communication (RS232 and RS485), and can communicate at up to 4.5 Mbps. UART0 (TXD0, RXD0, RST0 & CTS0 pins) can be used for communication. However, UART1 (TXD1 pin) features only data transmit signal so, it is usually used for printing log.

SPI Pins NodeMCU/ESP8266 features two SPIs (SPI and HSPI) in slave and master modes.

2.2.2 DHT11 Sensor

DHT11 is a sensor used to measure the temperature and humidity of the surrounding. DHT11 sensor comprises a capacitive humidity sensing element which is used to measure the humidity and an NTC thermistor for sensing the temperature. The humidity sensing capacitive element has two electrodes with a moisture-holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels.

The circuit IC measures and processes these resistance values and changes them into a measurable digital signal.

Features:

Operating Voltage: 3.3V-5.5V

Temperature Range: 0-50 °C

Temperature Accuracy: ±2% °C

Humidity Range: 20-90% RH

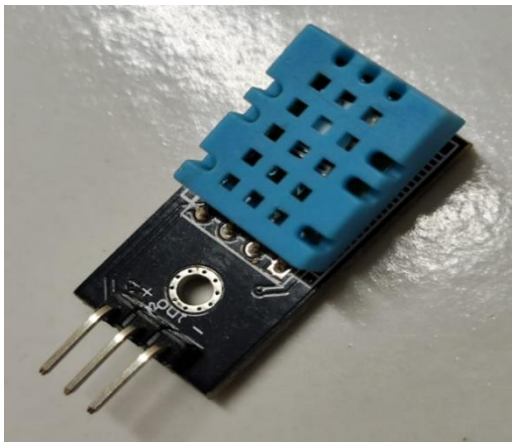


FIG 2.2 DHT11 Sensor

Humidity Accuracy: ±5% RHThe temperature range of DHT11 is from 0 to 50 degree Celsius with a 2-degree accuracy. Humidity range of this sensor is from 20 to 80% with 5% accuracy. The sampling rate of this sensor is 1Hz .i.e. it gives one reading for every second. DHT11 is small in size with operating voltage from 3 to 5 volts. The maximum current used while meFasuring is 2.5mA.

2.2.3 BMP280 Sensor

The BMP280 is one of the most advanced environmental sensors currently available on the market. It monitors temperature, pressure with great accuracy .with its tiny size, low power consumption and robust performance, this sensor makes a valuable tool for measuring atmospheric condition in any environment.

The BMP280 sensor is a high precision sensor module that detects atmospheric temperature, pressure. This module supports both the SPI interface as well as the I2C interface. Unlike

other sensors, these sensors have already been calibrated. They can start measuring temperature, pressure as soon as they are turned on and no further calibration is required.

As per the BMP280 datasheet, the sensor module requires a minimum voltage(VCC) of 1.71V, whereas sensor modules from earlier versions require 1.8V(VCC)

The BMP280 is an absolute barometric pressure sensor, which is suitable for mobile applications. It is very popular for its very compact dimensions and low power consumption. It is used in mobiles, smartwatches, weather stations, etc. The BMP280 is based on Bosch's proven Piezo-resistive pressure sensor technology featuring high accuracy and linearity as well as long-term stability.

FIG 2.3 BMP280 Sensor

Features of BMP280:

Pressure Range: 300-1100 hPa

Temperature Range: -40-85°C

Average typical current consumption:3.4 µA @ 1 Hz

Interface: I2C and SPI

Operating voltage: 1.71 V – 3.6 V

2.2.4 Rain Sensor

A rain sensor is one kind of switching device which is used to detect the rainfall. It works like a switch and the working principle of this sensor is, whenever there is rain, the switch will be normally closed.

The rain sensor module/board is shown below. Basically, this board includes nickel coated lines and it works on the resistance principle. This sensor module permits to gauge moisture through analog output pins & it gives a digital output while moisture threshold surpasses.

This module is similar to the LM393 IC because it includes the electronic module as well as a PCB. Here PCB is used to collect the raindrops. When the rain falls on the board, then it creates a parallel resistance path to calculate through the operational amplifier.

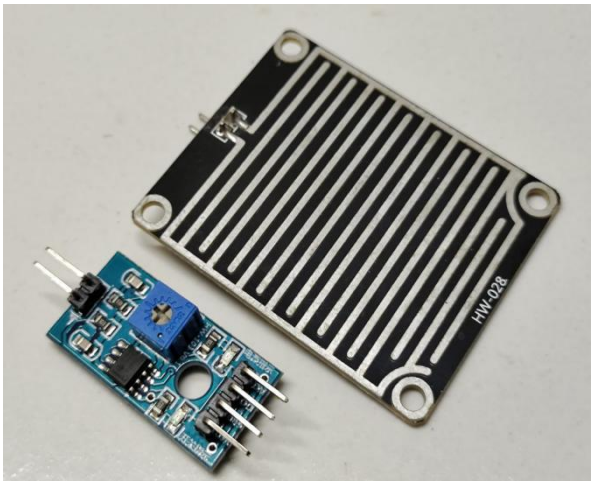


FIG 2.4 Rain Sensor

This sensor is a resistive dipole, and based on the moisture only it shows the resistance. For example, it shows more resistance when it is dry and shows less resistance when it is wet.

2.2.5 TP4056 Li-Ion Battery Charger Module

The TP4056 is a popular Lithium-Ion battery charger controller IC. It supports a constant current – constant voltage charging for a single cell 3.7v Li-Ion Battery. It comes with an 8-pin SOP package, which can be assembled with some external components to build a complete charging module.

The complete pin description of a TP4056 is shown below:"here just add the picture of it" . The TP4056 module is a small and affordable battery charger board. It's intended to charge single-cell lithium polymer or lithium ion batteries via USB or an external power source. The whole charging process is controlled by this module. This guarantees that the battery is charged effectively and safely without requiring a lot of additional parts or circuitry.

Pinout

-BAT+ and BAT-: Attach to the battery's positive and negative terminals.

-IN+ and IN-: Attach to the power supply's positive and negative terminals (typically 5V).

-OUT+ and OUT-: During battery charging, these pins are connected to a load that can receive electricity straight from the battery.

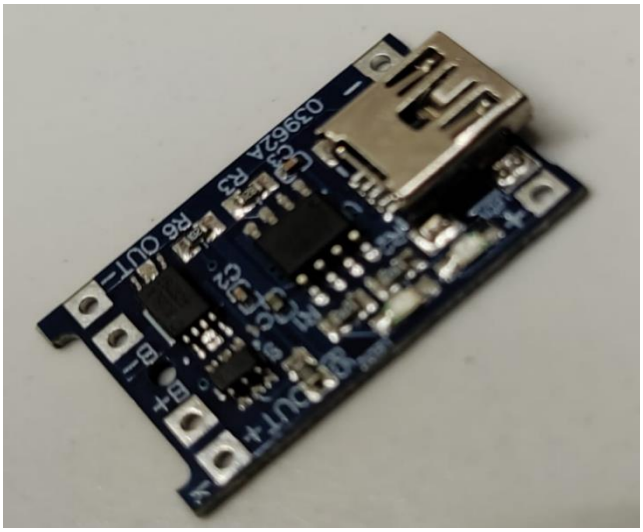


FIG 2.5 TP4056 Li-Ion Battery Charger Module

Applications of the TP4056 Module

-Portable electronics

-DIY battery packs

-Battery management systems

-Power banks

-Solar chargers

2.2.6 3.7V 18650 Li-ion Battery

A 3.7V 18650 Li-ion battery works by using lithium ions to move between the positive (cathode) and negative (anode) electrodes through an electrolyte, storing and releasing energy during charging and discharging, respectively.

The 18650 battery is a cylindrical cell with a diameter of 18mm and a length of 65mm. It consists of a cathode (positive electrode), an anode (negative electrode), an electrolyte, and a separator.

The cathode is typically made of a metal oxide, while the anode is made of a porous carbon material. During charging, an external power source applies a voltage greater than the cell's own voltage, forcing electrons to flow from the positive to the negative electrode. Simultaneously, lithium ions move from the cathode to theanode through the electrolyte and separator, becoming embedded in the porous anode material

When the battery is used, the process reverses. Lithium ions move from the anode to the cathode through the electrolyte and separator, releasing electrons and creating an electrical current.

FIG 2.6 3.7V 18650 Li-ion Battery

The nominal voltage of an 18650 battery is 3.7V, but the voltage can range from 3.0V when fully discharged to 4.2V when fully charged. 18650 batteries typically have a capacity of 2500mAh to 3500mAh.

18650 batteries are widely used in various devices, including flashlights, laptops, electric vehicles, and power tools. Their high energy density and reliability make them a popular choice for portable electronic devices.

2.2.7 DC-DC 0.9V-5V Step-Up Booster

This module is perfect for creating a portable charger since it can transform 0.9-5V input into 5V output for most digital devices. It can also be utilized for USB charger projects and onboard USB device supply solutions, including GoPro cameras.

You can drive it with a single-cell LiPO, AA battery, or other. With two AA batteries, it can deliver 500 ~ 600MA of current, and with one AA battery, it can deliver 200ma of current. This is ideal for powering mobile phones, cameras, microcontrollers, and other digital devices.

A DC-DC Step-Up Booster (or boost converter) is a power supply that takes a lower input voltage and pumps it up to a higher output voltage. On your example, it takes an input voltage of 0.9V and outputs 5V. This is especially handy when you have a low-voltage source (such as a battery) but require higher voltage for powering some devices.

FIG 2.7 DC-DC 0.9V-5V Step-Up Booster

The inductor is the heart of the boost converter, which stores energy temporarily. During operation of the converter, current passes through the inductor, generating a magnetic field.

The switch, most often a MOSFET or transistor, rapidly opens and closes (in the kilohertz range) to manage the transfer of energy. When the switch is closed, current flows

from the input through to the inductor, and the magnetic field in the inductor accumulates. When the switch is open, the energy stored in the inductor is released.

2.2.8 Solar Panel 6V, 100Ma

This is a high-efficiency solar panel output of 100 mA of current at 6V in perfect light conditions. It is a Durable anti-eye solar panel with superior performance in weak sunlight as well. Besides, a Unique technology adopted to avoid water freeze deformation frame with an Elegant picture Small portable size and convenient carry.

Solar Panels devices that are used to absorb the sun's rays and convert them into electricity or heat. A solar panel really an assembly of photovoltaic cells, employed to create electricity from the photovoltaic effect. Several photovoltaic cells mounted in series form a solar panel. In which the photons, or light particles, knocks the electrons loose from the atoms, producing a stream of electricity. A photovoltaic cell essentially a sandwich composed of two slices of semi-conducting material, typically silicon.

FIG 2.8 Solar Panel 6V, 100M

The 6V 100mA small solar panel commonly used for our development boards with an LCD in reasonable sunlight conditions. This particular panel specifically designed to charge 3.6v AA batteries. At the same time, this 6V 100mA Solar Panel works with 36-cell Solar Panel sized 70*70 cm without the need for a frame or any special altering. Additionally, these Polycrystalline mini solar cells are Laser cut to the correct size and

encapsulated in the special sun and weather-resistant durable outer poly frame or injection molded trays custom designed giving them distinctive features.

2.2.9 Jumper wires

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed.

Though jumper wires come in a variety of colors, the colors don't actually mean anything. This means that a red jumper wire is technically the same as a black one. But the colors can be used to your advantage in order to differentiate between types of connections, such as ground or power.

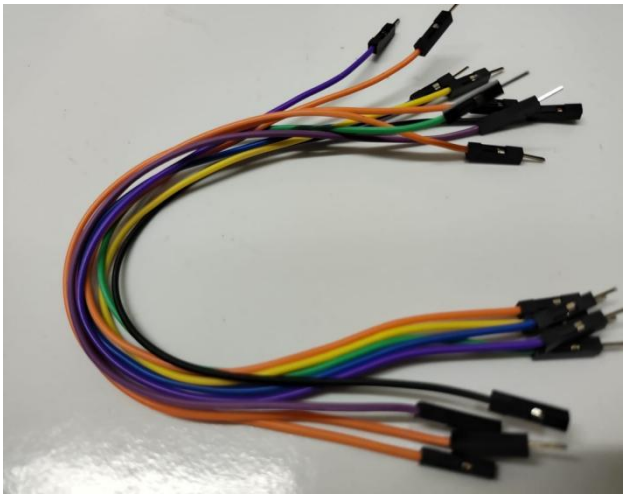


FIG 2.9 Jumper wires

Jumper wires typically come in three versions: male-to-male, male-to-female and female-to-female. The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things, while female ends do not and are used to plug things into. Male-to-male jumper wires are the most common and what you likely will use most often. When connecting two ports on a breadboard, a male-to-male wire is what you'll need.

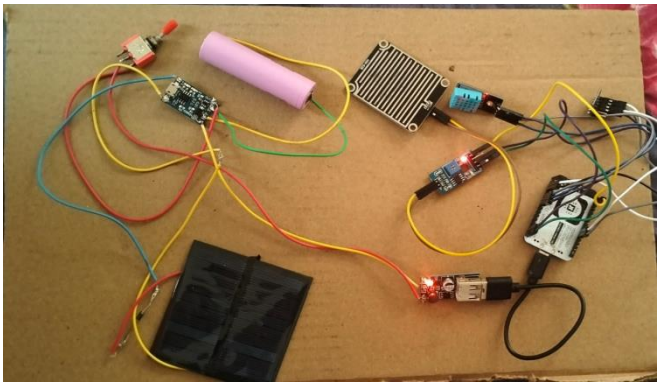


FIG 2.10 Circuit Connection

CHAPTER-3

PROGRAMMING

3.1 Explanation of the code

Self Powered Wi-Fi Weather Station using NodeMCU and ThingSpeak

This code is an Arduino sketch designed to read environmental data from various sensors and send that data to ThingSpeak, a cloud-based IoT platform. It utilizes an ESP8266 microcontroller to connect to Wi-Fi and communicate with the sensors. The sketch includes several libraries: Wire.h for I2C communication necessary for the BMP280 sensor, Adafruit_Sensor.h for a unified interface for various sensors, Adafruit_BMP280.h specifically for the BMP280 sensor that measures temperature and pressure, DHT.h for the DHT11 temperature and humidity sensor, ESP8266WiFi.h for connecting the ESP8266 to Wi-Fi networks, and ThingSpeak.h for sending data to the ThingSpeak platform.

The code begins by defining the Wi-Fi credentials, storing the SSID and password for the network that the ESP8266 will connect to. It also includes the ThingSpeak channel details, which consist of a channel ID and an API key for writing data to a specific channel. Sensor objects are created for the BMP280 and DHT11 sensors, with the DHT11 connected to GPIO pin D4. Additionally, the rain sensor is set up using an analog pin (A0) to read its values.

3.2 Final Code

```
#include <Wire.h>

#include <Adafruit_Sensor.h>

#include <Adafruit_BMP280.h>

#include <DHT.h>

#include <ESP8266WiFi.h>

#include <ThingSpeak.h>



sss// Wi-Fi credentials

const char* ssid = "Empty";
```

```
const char* password = "lakshmi1272";


// ThingSpeak channel details
unsigned long channelID = 2888912;
const char* writeAPIKey = "KSNEHKSHJOPIGEZ1";


// Create sensor objects
Adafruit_BMP280 bmp;
DHT dht(D4, DHT11); // DHT11 sensor connected to GPIO2 (D4)


// Rain sensor setup
#define RAIN_SENSOR_PIN A0 // Analog pin connected to AO of the rain sensor


WiFiClient client;


void setup() {
  Serial.begin(115200);


  // Connect to Wi-Fi
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
```

```
  }
  Serial.println("Connected to WiFi");

  // Initialize BMP280 sensor
  if (!bmp.begin(0x76)) {
    Serial.println("Could not find a valid BMP280 sensor, check wiring!");
    while (1);
  }

  // Initialize DHT11 sensor
  dht.begin();

  // Initialize ThingSpeak
  ThingSpeak.begin(client);
}

void loop() {
  // Read BMP280 data
  float temperatureBMP = bmp.readTemperature();
  float pressureBMP = bmp.readPressure() / 100.0F; // Pressure in hPa

  // Read DHT11 data
  delay(2000); // DHT11 needs at least 2 seconds between reads
```

```
float humidityDHT = dht.readHumidity();

float temperatureDHT = dht.readTemperature();


// Read Rain sensor data

int rainValue = analogRead(RAIN_SENSOR_PIN);


// Print sensor values to Serial Monitor

Serial.print("BMP280 Temperature: ");

Serial.print(temperatureBMP);

Serial.println(" *C");

Serial.print("BMP280 Pressure: ");

Serial.print(pressureBMP);

Serial.println(" hPa");

Serial.print("DHT11 Humidity: ");

Serial.print(humidityDHT);

Serial.println(" %");

Serial.print("DHT11 Temperature: ");

Serial.print(temperatureDHT);

Serial.println(" *C");

Serial.print("Rain Sensor Value: ");

Serial.println(rainValue);


// Update ThingSpeak fields
```

```
  ThingSpeak.setField(1, temperatureBMP);    // Field 1: BMP280 Temperature
  ThingSpeak.setField(2, pressureBMP);       // Field 2: BMP280 Pressure
  ThingSpeak.setField(3, humidityDHT);       // Field 3: DHT11 Humidity
  ThingSpeak.setField(4, temperatureDHT);    // Field 4: DHT11 Temperature
  ThingSpeak.setField(5, rainValue);         // Field 5: Rain Sensor Value

  // Write data to ThingSpeak
  int responseCode = ThingSpeak.writeFields(channelID, writeAPIKey);
  if (responseCode == 200) {
    Serial.println("Data successfully sent to ThingSpeak");
  } else {
    Serial.print("Error sending data to ThingSpeak. Response code: ");
    Serial.println(responseCode);
  }

  // Wait before next reading
  delay(20000); // ThingSpeak allows updates every 15 seconds
}
```

CHAPTER 4

RESULTS

4.1 Results

The solar weather monitoring system represents a paradigm shift in environmental observation, leveraging cutting-edge technology to provide real-time insights into weather patterns. By harnessing solar energy and integrating advanced sensors, this system offers a sustainable and efficient solution for monitoring temperature, humidity, atmospheric pressure, and precipitation.

Theoretical Framework

1. Systems Thinking: The solar weather monitoring system embodies a holistic approach, integrating multiple components to provide a comprehensive understanding of weather dynamics.

2. Energy Harvesting: The system's reliance on solar energy underscores the importance of renewable energy sources in powering environmental monitoring initiatives.

3. Data-Driven Decision-Making: The system's real-time data transmission capabilities facilitate informed decision-making in various industries, from agriculture to emergency management.

Key Implications

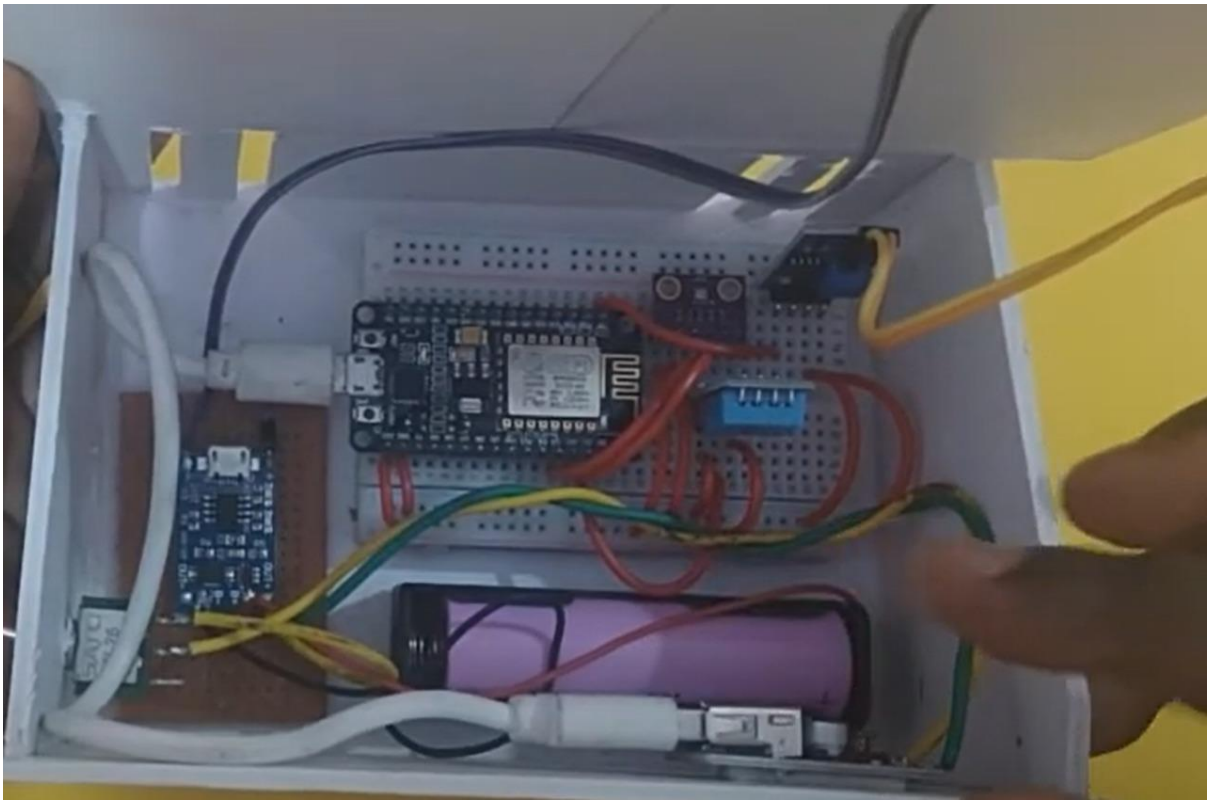
1. Enhanced Predictive Capabilities: The system's accurate and reliable data can improve weather forecasting models, enabling more effective planning and preparedness.

2. Increased Efficiency: Automation and real-time data transmission can streamline weather monitoring tasks, reducing costs and improving resource allocation.

3. Environmental Stewardship: The system's solar-powered design promotes sustainability and reduces carbon footprint, aligning with environmental conservation goals.

Future Directions

1. Integration with Emerging Technologies: Exploring the potential of integrating the solar weather monitoring system with emerging technologies, such as IoT devices or AI-powered analytics.



4.1 *Self Powered Weather Monitoring System*

2. Scalability and Adaptability: Developing strategies to scale the system for larger applications and adapt it to diverse environmental contexts.

3. Interdisciplinary Collaborations: Fostering collaborations between environmental scientists, engineers, and policymakers to ensure the system's effective deployment and impact.

4.2 Summary Of The Project

A solar weather monitoring system using NodeMCU and ThingSpeak is designed to collect and transmit real-time weather data, such as temperature, humidity, and wind speed, via Wi-Fi. This system operates on solar power, ensuring sustainability and continuous data logging for analysis and visualization on the ThingSpeak platform. The core of the system is the NodeMCU ESP8266, a Wi-Fi enabled microcontroller that facilitates internet connectivity and data transmission. It is equipped with various sensors, including the DHT22 for measuring temperature and humidity, the BMP180 for atmospheric pressure and altitude, a rain sensor for detecting rainfall, and an LDR (Light Dependent Resistor) for

measuring light intensity. The power supply consists of a solar panel and a rechargeable Li-ion battery (e.g., 18650), ensuring continuous operation.

The functionality of the system includes data collection from the sensors at regular intervals and data transmission to the ThingSpeak cloud platform for storage and analysis, allowing users to access real-time weather information from anywhere. Additionally, the NodeMCU can enter a low-power deep sleep mode to conserve battery life, waking up periodically to take measurements. The benefits of this system include remote monitoring capabilities, enabling users to track weather conditions via a web interface or mobile app, as well as cost-effectiveness due to the use of solar power and affordable components, making it accessible for personal and educational projects. Furthermore, the system is scalable, allowing for the addition of more sensors or nodes to cover larger areas or monitor additional parameters.

Reference

The references for the Solar Weather Monitoring System project using the NodeMCU (ESP8266) microcontroller encompass a variety of essential resources for its development. The official documentation provided by Espressif Systems offers detailed insights into the hardware specifications, programming interfaces, and development tools specific to the ESP8266 microcontroller, which is crucial for understanding the capabilities and limitations of the NodeMCU board. Additionally, community forums and discussion boards, such as the Arduino forum, ESP8266 community forums, and Stack Overflow, serve as valuable resources for seeking assistance, sharing project ideas, and accessing code snippets and tutorials related to the NodeMCU and weather monitoring systems.

https://iotdesignpro.com/projects/solar-powered-wi-fi-weather-station-using-nodemcu-and-thingspeak